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May-25-1098

MCA-6202 (Java Programming)

MCA 2nd (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from Sections A, B, C and D. Section E is compulsory.

SECTION - A

1. What is the difference between a method and a constructor? Explain the concept and implementation of method overloading in JAVA. (12)

OR

2. Explain JDK, JRE, and JVM in detail. How does JVM make Java platform-independent? (12)

SECTION - B

3. What do you mean by multithreading? Write a JAVA program to illustrate the use of multithreading. (12)

OR

4. Write a program that takes an integer input from the user, checks if the number is valid (positive integer), and performs division by a fixed number (e.g., 10). If the input is invalid (non-integer or negative), the program should throw an exception and handle it using try-catch-finally. (12)

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SECTION - C

5. What is AWT? How is it different from Swing? Explain key GUI components of AWT. (12)

OR

6. Write a Swing program that have 5 text fields and one submit button. When you press the button first 4 text field's average has to be displayed in the 5th text field. (12)

SECTION - D

7. What is JDBC? Write a step-by-step guide on how to establish a database connection using JDBC. (12)

OR

8. Explain the concept of event handling in Java. Discuss different event classes and listener interfaces. (12)

SECTION - E (Compulsory)

9. Attempt ALL questions from this section. Each question carries 2 marks.

- (i) Explain OOPs concept in JAVA.
- (ii) Explain multilevel inheritance with an example in Java.
- (iii) Describe the life cycle of thread.
- (iv) What is the Java Virtual Machine (JVM)?
- (v) What is the difference between String, StringBuffer in JAVA?
- (vi) Explain different data types used in Java. (5×2=12)